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## **Voices from the Pedal Harp Lab**

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A long-form feature on the engineers, makers, and performers rebuilding orchestral harp practice from first principles.

*By Annika Voss – Photographs by Esther Tan*

When Eleanor Whitcombe first walked into the workshop in the autumn of 2023, she was carrying two things: a battered hardshell case containing the prototype of what she would later call the *Lyrebird Concert Mark IV*, and a dog-eared spiral notebook full of measurements that she had been adding to since her postdoc at Stanford. The case went on a workbench under a north-facing skylight. The notebook went into a drawer. Two years later, the prototype is the production instrument that more than thirty professional harpists now use, and the notebook – mostly forgotten in that drawer – has become a quiet bestseller within a small community of acoustic researchers who trade scanned pages on a private mailing list.

This is the story of a small instrument and the people who keep remaking it. It is also the story of why, exactly, the pedal harp has resisted the standard innovations that swept through every other orchestral string family in the twentieth century – and what happens when a generation of engineers, performers, and luthiers finally decides that the resistance has gone on long enough.

## **A workshop, and a problem**

The *Lyrebird* workshop occupies the second floor of a converted button factory in Cambridge, Massachusetts. From the street it is un-

remarkable: brick, two stories, a small brass plate by the door that reads *Lyrebird Acoustics, Inc.* in a font so understated that half the visitors I observed walked past it twice. Inside, the smell is of cedar shavings and freshly cut spruce, with an under-note of hide glue warming on a hot plate. The workshop is open to the public on Wednesday afternoons; on the day I visited there were two high-school students and an emeritus professor of art history sitting in folding chairs by the tuning bench, watching Whitcombe voice a row of newly strung wires.

*"The whole instrument was designed in 1810, and almost nothing has changed since," Whitcombe told me, plucking a string with the side of her thumbnail. "If you handed an Erard pedal harp from 1815 to a player today, they could tune it up and play a recital that afternoon. That is amazing, but it is also a problem."*

The problem, in Whitcombe's telling, is that "almost nothing has changed" actually means "almost nothing was *measured*". The pedal harp's neck-to-body angle, the soundboard taper, the string diameters, even the action of the pedal mechanism – all of these were settled by trial, error, and the conservative tastes of a small group of nineteenth-century makers, and then frozen in place by the prestige of the instruments those makers produced. There is no modern body of measurement that says *the soundboard should be 3.2 millimetres at the bass end and 1.6 at the treble*; there are only the instruments themselves, and the centuries of imitation that followed.

!!! note "What "voicing" means" *Voicing* is the luthier's term for adjusting an instrument after it is built so that each string speaks evenly and the tonal colour is consistent across the range. On a harp, voicing is mostly a matter of micro-adjusting the soundboard tension and the position of the *pin-block* (the wooden bar that holds the tuning pegs). A skilled voicer can

spend two full days on a single instrument; an unskilled one can spend two minutes.

## The notebook

The drawer-bound notebook contains, by my rough count, about eleven hundred individual measurements. Soundboard impedance plots taken at sixteen different points on twenty different instruments. String break-angle photographs in dozens of frames per instrument. Pedal action force-curves – the relationship between the player's foot pressure and the resulting pitch change – recorded as ASCII text files dumped from a sensor rig that Whitcombe and her then-supervisor built out of a kitchen scale and a piece of bicycle inner tube.

Most of the notebook is, by Whitcombe's own admission, "noise". A particular string sounded slightly sharper than expected; a particular soundboard had a resonance at 380 Hz that no other soundboard in the sample had. Treated as individual data points the entries are not especially illuminating. But aggregated – and Whitcombe spent three months in the winter of 2022 aggregating them, on a borrowed laptop in her childhood bedroom in Salem, Oregon – they begin to tell a story. The story is that the harps of the pedal-harp tradition are each idiosyncratic in *small* ways, but extremely consistent in *large* ones. The neck-to-body angle is essentially identical across the sample. The soundboard taper falls into one of two distinct patterns. The string-tension profile – pounds of tension per millimetre of string diameter, plotted against pitch – is so similar across the twenty instruments measured that the curves overlay each other within the thickness of a pencil line.

*"It was the consistency that surprised me," Whitcombe said. "I went into the project assuming that I would find these huge variations between makers, and that the variations would explain why some harps sound the way they do and others*

*sound differently. What I found instead was the opposite. The harps that sound very different have almost the same measurements. The differences are all in the last one percent."*

## Why a feature article?

I wrote this article because the pedal harp is an instrument that hides in plain sight. It appears in every major orchestral score written after 1860, contributes to dozens of film scores per year, and supports a small but devoted community of soloists. It is also the most physically complex single instrument an orchestra owns: a modern concert harp has 47 strings, seven double-action pedals, a linkage system involving more than two thousand individual components, and a body weighing close to ninety pounds.

The instrument is too important to leave to its own conservatism. What Whitcombe and her colleagues are doing in the Cambridge workshop – and what a handful of other workshops in Italy, Germany, and Japan are doing in parallel – is the slow, patient work of turning a craft into a discipline. They are measuring what was never measured. They are publishing what was never published. They are producing the kind of *quantitative ground truth* that every other orchestral instrument family acquired in the second half of the twentieth century, and that the pedal harp, alone among the strings, never did.

## The next generation

I spent three days at the Lyrebird workshop. I met the apprentice Marcus Ito, who is twenty-two and learning the trade after a computer-science degree and two years at a robotics startup. I met the senior voicer Yuki Tanaka, who came over from a workshop in Saita-

ma in 2019 and has the calmest hands I have ever watched at a workbench. I met the cellist-turned-acoustician Henri Fabre, who is on sabbatical from a Paris conservatory and whose laptop is covered with stickers from open-source acoustic software projects. I sat in on three voicing sessions, two tuning sessions, and one extended conversation between Whitcombe and Tanaka about whether the *nineteenth* pedal coupling – a notoriously balky link in the double-action mechanism – could be replaced with a tensioned cable instead of the traditional metal rod.

*"I want my children to inherit an instrument that has been understood," Whitcombe said on my last afternoon, putting the Mark IV back in its case after a recital-pitch tuning. "Not just an instrument that has been played."*

It is too early to say whether she will succeed. The Lyrebird Mark IV is now in the hands of thirty-three professional harpists; the acoustic-measurements project that grew out of the spiral notebook has been adopted, in part, by the Conservatoire National Supérieur de Musique in Paris. But the pedal harp is a slow instrument, and the traditions it embodies are slow traditions. Whatever the generation now in the Lyrebird workshop produces, it will take fifty years to be sure.

In the meantime there is the work. There are the measurements. There is the small community of luthiers and engineers and players who trade scanned notebook pages on a private mailing list, and who – when I asked Whitcombe whether the list had a name – told me, with a small smile, that of course it did, but that she would prefer I not print it. Some things, she said, are still small enough to want to stay small.

- *The Modern Concert Harp*, by Marisol Aguilar. The standard textbook on twentieth-century harp making.
- *Acoustic Measurements for Luthiers*, by Henri Fabre. The book that grew out of Fabre's open-source software project.
- *The Pedal Harp: An Encyclopedia*, by various authors. A comprehensive reference, in five volumes.
- *Strings Attached*, a documentary film by Cassie Yu (2024). A 90-minute portrait of three harpists at three different points in their careers.

**+4p Annika Voss**

writes about acoustic instruments and the people who build them. Her previous feature for this magazine, *The Last of the Cellists*, appeared in Issue 02.

## A reading list